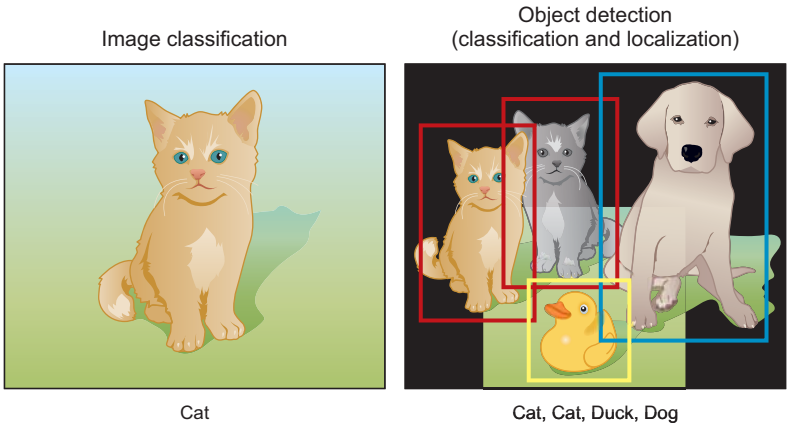

Figure 7.1 Image classification vs. object detection tasks. In classification tasks, the classifier outputs the class probability (cat), whereas in object detection tasks, the detector outputs the bounding box coordinates that localize the detected objects (four boxes in this example) and their predicted classes (two cats, one duck, and one dog).

Table 7.1 Image classification vs. object detection

Image classification	Object detection
The goal is to predict the type or class of an object in an image. ▪ Input: an image with a single object ▪ Output: a class label (cat, dog, etc.) ▪ Example output: class probability (for example, 84% cat)	The goal is to predict the location of objects in an image via bounding boxes and the classes of the located objects. ▪ Input: an image with one or more objects ▪ Output: one or more bounding boxes (defined by coordinates) and a class label for each bounding box ▪ Example output for an image with two objects: – box1 coordinates (x, y, w, h) and class probability – box2 coordinates and class probability Note that the image coordinates (x, y, w, h) are as follows: (x and y) are the coordinates of the bounding-box center point, and (w and h) are the width and height of the box.

fits the detected object. This is a challenging CV task because it requires both successful object localization, in order to locate and draw a bounding box around each object in an image, and object classification to predict the correct class of object that was localized.

Object detection is widely used in many fields. For example, in self-driving technology, we need to plan routes by identifying the locations of vehicles, pedestrians, roads, and obstacles in a captured video image. Robots often perform this type of task to